

Departure Delay Analysis

DATA 5300, Group Project #1

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Agenda

Introduction

- **Problem Statement**
- Technique
- Sources
- Timeline of Analysis

RA

- Initial Preparation
- Summary

EDA

- Variables
- Correlation
- Summary

Permutation Testing

- Hypothesis 1
- Hypothesis 2

Discussion

- Key Takeaways
- Potential Expansion

Introduction

- Problem Statement
- Technique
- Data Sources
- Analytic Approach

Introduction



Problem Statement:

What variable has the greatest effect on departure delays?

Libraries

Link to Google Colab Notebook: [departure_delay_analysis.ipynb](#)

Packages

- **'nycflights13'**
 - *Weather*
 - *Airlines*
 - *Airports*
 - *Flights*

Libraries

- **Ggplot2**: Imported for data visualization
- **Dplyr**: Used for data manipulation in R. Such as the filter, select, mutate, arrange, and summarize functions
- **Tidyr**: Used for tidying and reshaping data
- **Tibble**: Installed modern data frames for R
- **Forcats**: Imported for categorical/factor data manipulation
- **Lubridate**: Used to work with date and time

Technique

Airline: United Airlines

Year: 2013

Location: NYC

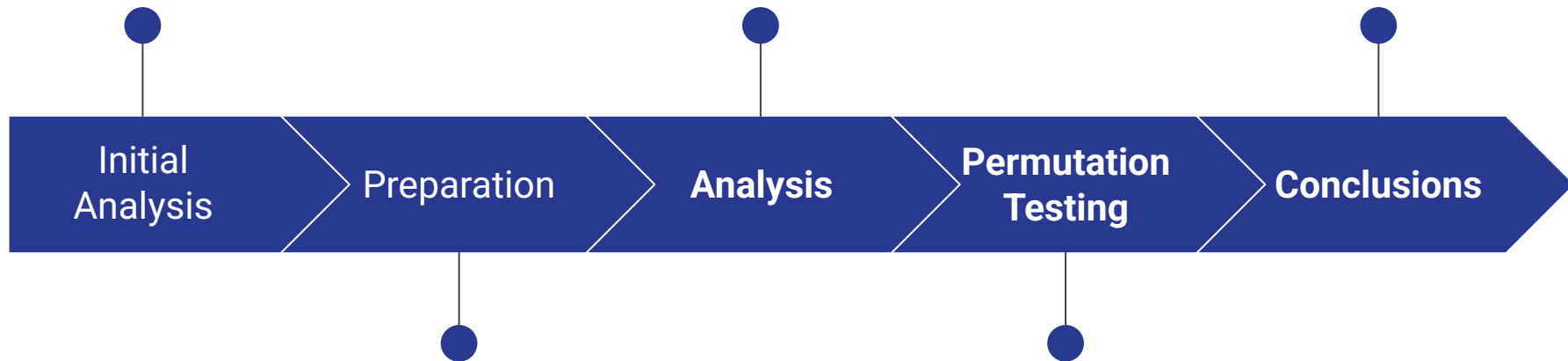
Analysis Variables:

- Time of day
- Time of year
- Temperature
- Wind speed
- Precipitation
- Visibility

Exploring the data sets
as they were, without
changes

Perform the Regression
Analysis and Exploratory
Data Analysis

Summary of Findings



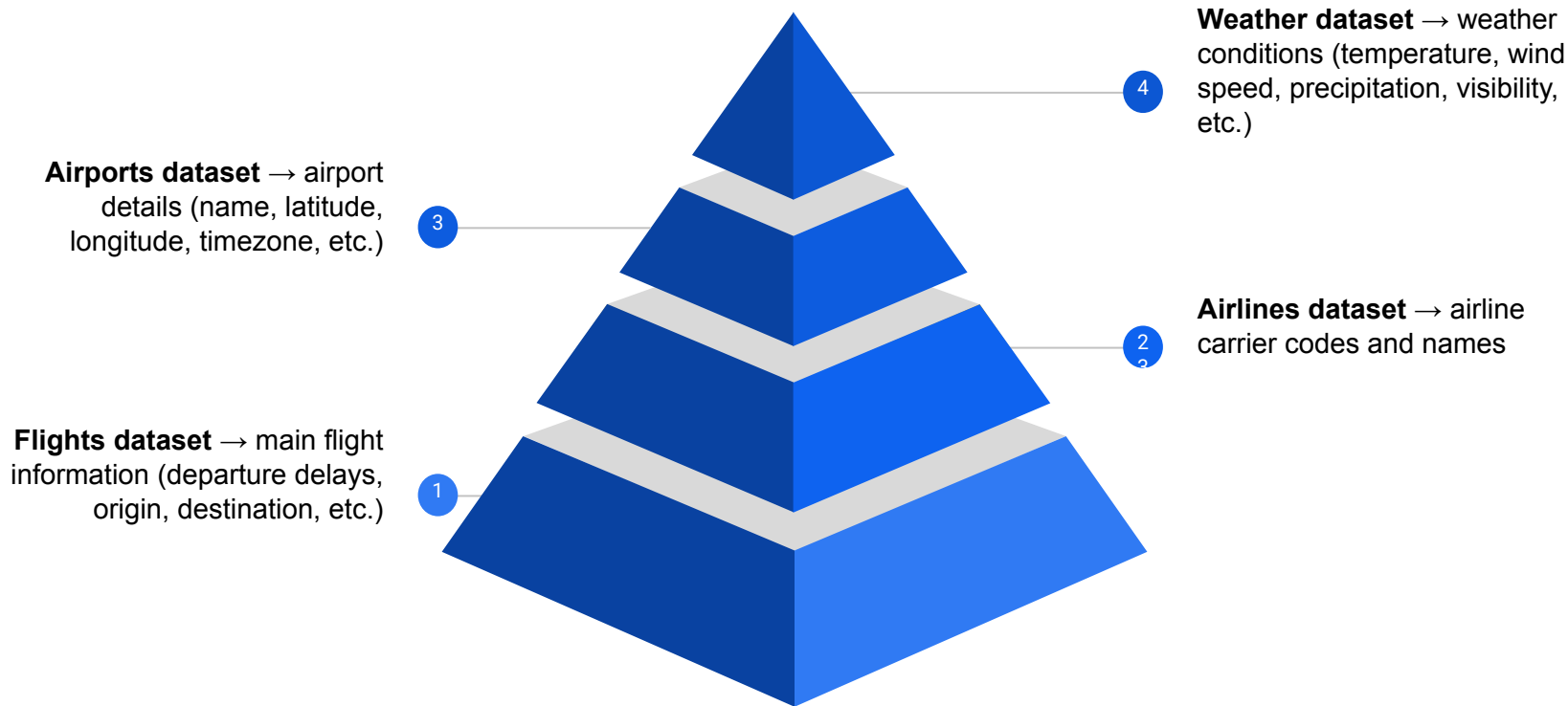
Clean, prep, merge and
filter the data sets to
ONE clean set of values

Test our final
hypothesis for
randomization

Regression Analysis

- Initial Data Preparation
- Summary

Data Preparation: Merging the Data



Multiple Regression Analysis

Residuals:

Min	1Q	Median	3Q	Max
-39.96	-15.09	-8.71	-0.47	470.21

Coefficients: (2 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-16.38606	3.00660	-5.450	5.04e-08	***
temperature	0.38169	0.03883	9.829	< 2e-16	***
wind_speed	-0.15011	0.05012	-2.995	0.00274	**
precipitation	NA	NA	NA	NA	
visibility	NA	NA	NA	NA	
time_of_dayEvening	8.64888	0.19085	45.317	< 2e-16	***
time_of_dayMorning	-8.32353	0.17275	-48.181	< 2e-16	***
time_of_dayNight	7.80080	0.51586	15.122	< 2e-16	***
time_of_yearSpring	16.00571	1.64031	9.758	< 2e-16	***
time_of_yearSummer	0.90259	0.22690	3.978	6.96e-05	***
time_of_yearWinter	16.14118	1.67800	9.619	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 34.8 on 234651 degrees of freedom
Multiple R-squared: 0.03919, Adjusted R-squared: 0.03915
F-statistic: 1196 on 8 and 234651 DF, p-value: < 2.2e-16

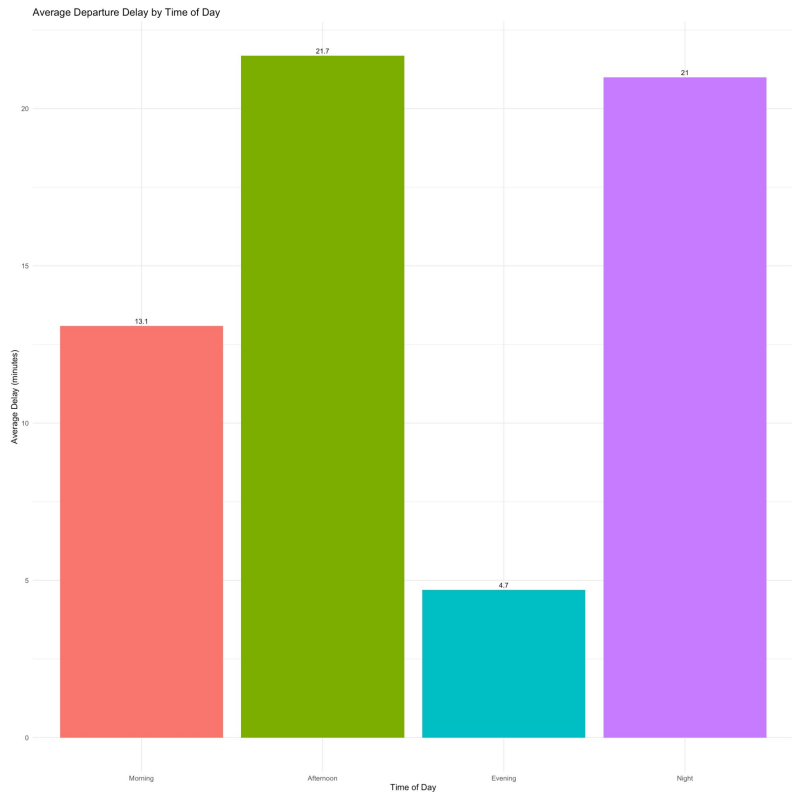
Adjusted R squared value: 0.039

Exploratory Data Analysis

- Variables
- Correlation
- Descriptive Analysis (visuals)

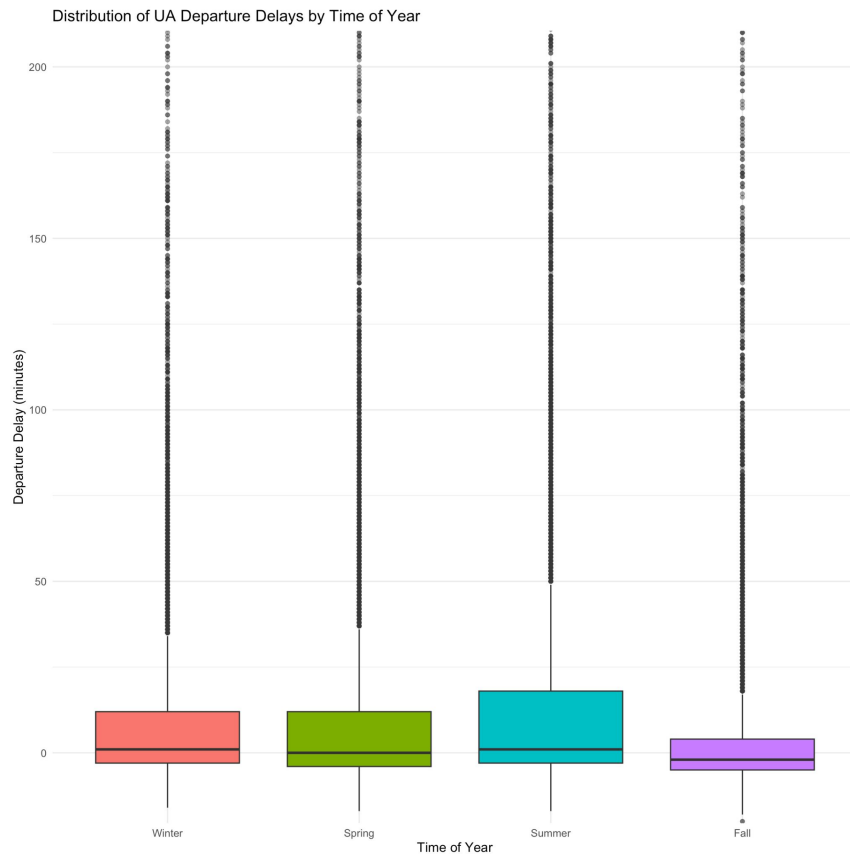
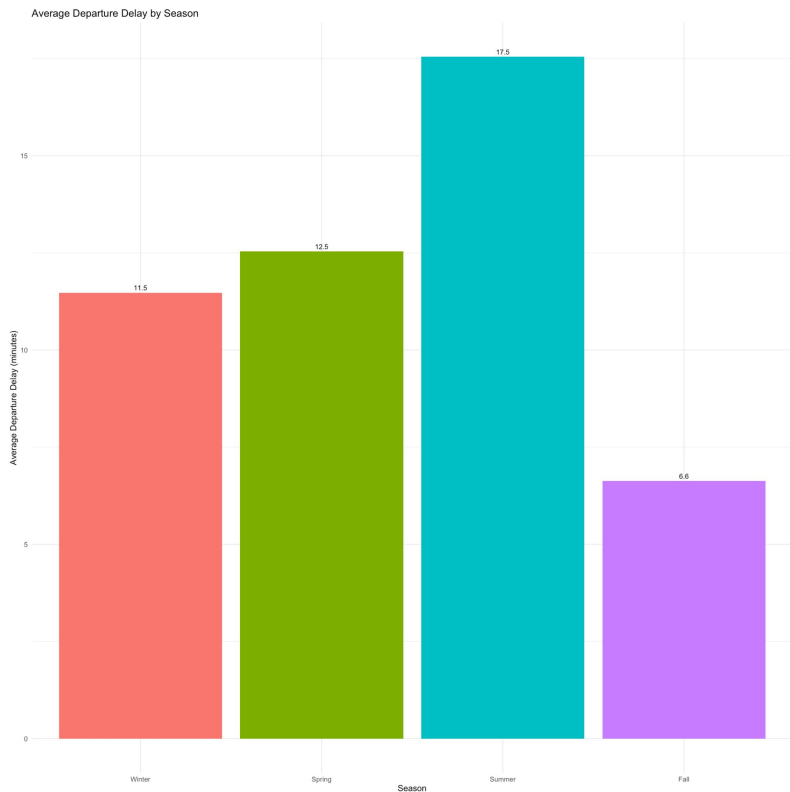
Exploratory Data Analysis

Average departure delay vs time of day



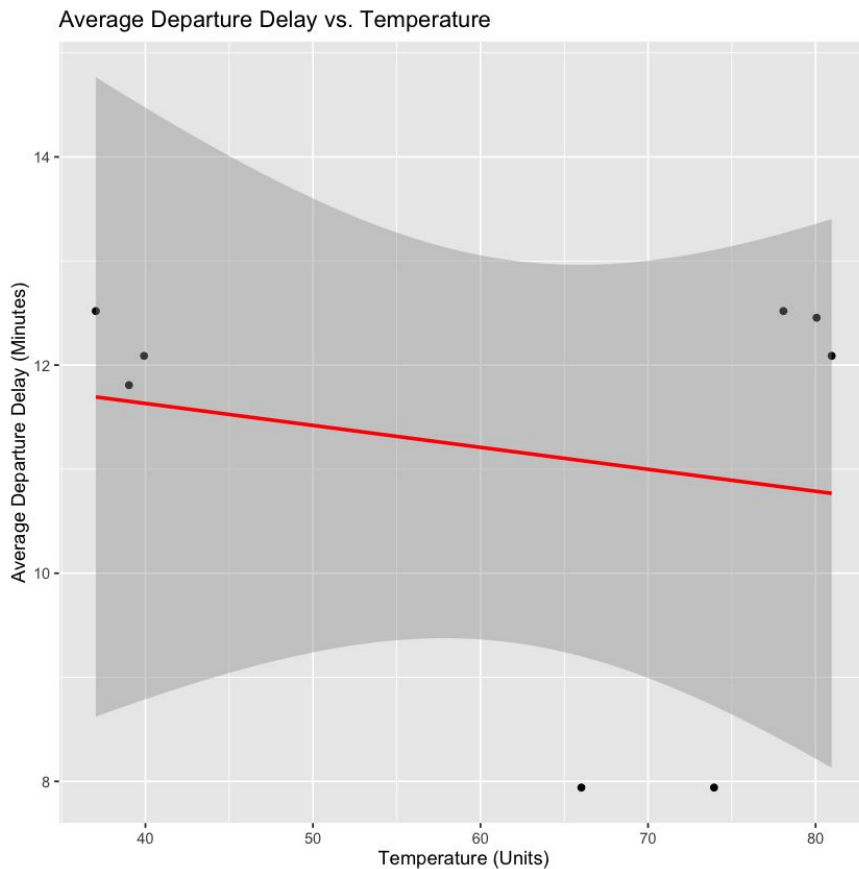
Exploratory Data Analysis

Average departure delay vs time of year



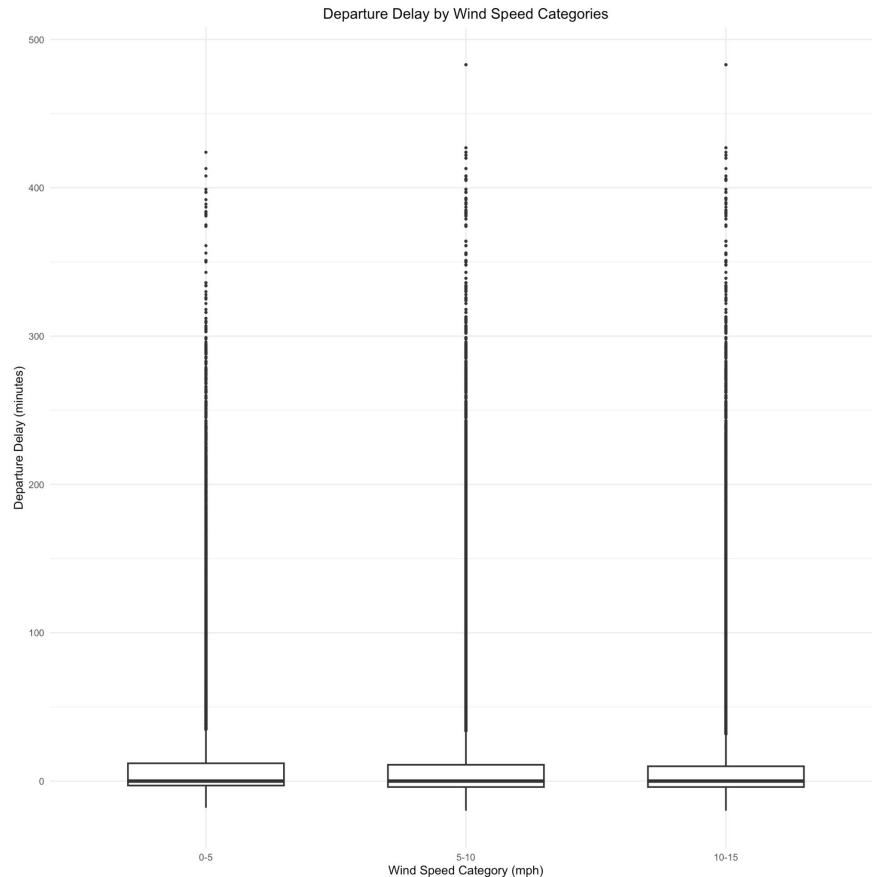
Exploratory Data Analysis

Average departure delay vs
Temperature



Exploratory Data Analysis

Average departure delay
vs WindsSpeed

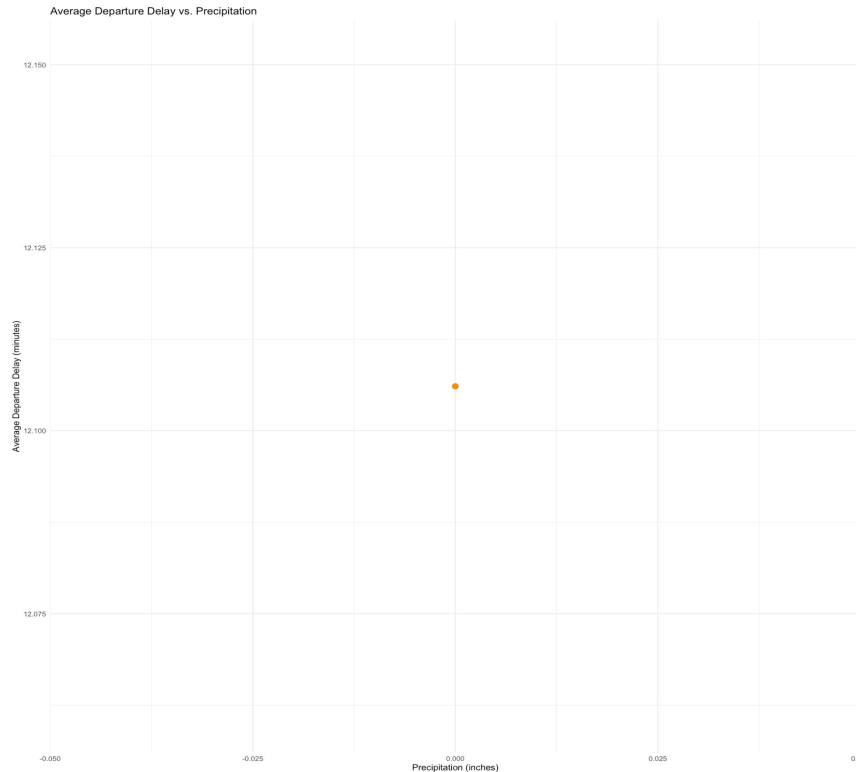


Exploratory Data Analysis

Average departure delay vs Precipitation

A tibble: 1 x 7

count	mean	median	min	max	sd	p95
<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
234660	0	0	0	0	0	0



Permutation Testing

- Hypothesis 1
- Hypothesis 2

Hypothesis 1

Does **time of day** affect departure delays?

```
library(dplyr)
library(infer)

# Filter data and ensure no missing values
ua_timeofday <- ua_combined %>%
  filter(!is.na(time_of_day), !is.na(departure_delay_minutes))

# Observed F-statistic, variation in delays (since time_of_day has >2 groups)
obs_stat_day <- ua_timeofday %>%
  specify(departure_delay_minutes ~ time_of_day) %>%
  calculate(stat = "F")

# Permutation test (randomizing time_of_day labels)
# shuffle 500 times and recalculate
set.seed(123)
perm_day <- ua_timeofday %>%
  specify(departure_delay_minutes ~ time_of_day) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 500, type = "permute") %>%
  calculate(stat = "F")

# Compute p-value
p_value_day <- perm_day %>%
  get_p_value(obs_stat = obs_stat_day, direction = "greater")

p_value_day
```

A tibble: 1

× 1

p_value

<dbl>

0

Hypothesis 2

Does **time of year** affect departure delays?

```
# Filter for complete data
ua_timeofyear <- ua_combined %>%
  filter(!is.na(time_of_year), !is.na(departure_delay_minutes))

# Observed F-statistic
obs_stat_year <- ua_timeofyear %>%
  specify(departure_delay_minutes ~ time_of_year) %>%
  calculate(stat = "F")

# Permutation test (randomizing time_of_year labels)
set.seed(123)
perm_year <- ua_timeofyear %>%
  specify(departure_delay_minutes ~ time_of_year) %>%
  hypothesize(null = "independence") %>%
  generate(reps = 500, type = "permute") %>%
  calculate(stat = "F")

# Compute p-value
p_value_year <- perm_year %>%
  get_p_value(obs_stat = obs_stat_year, direction = "greater")

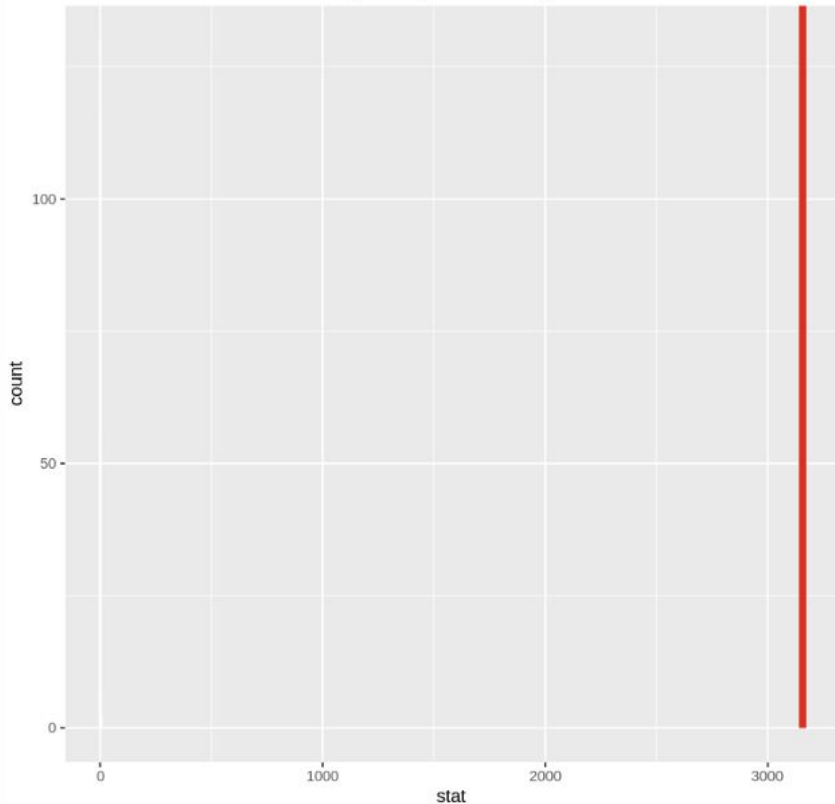
p_value_year
```

A tibble: 1
× 1
p_value
<dbl>

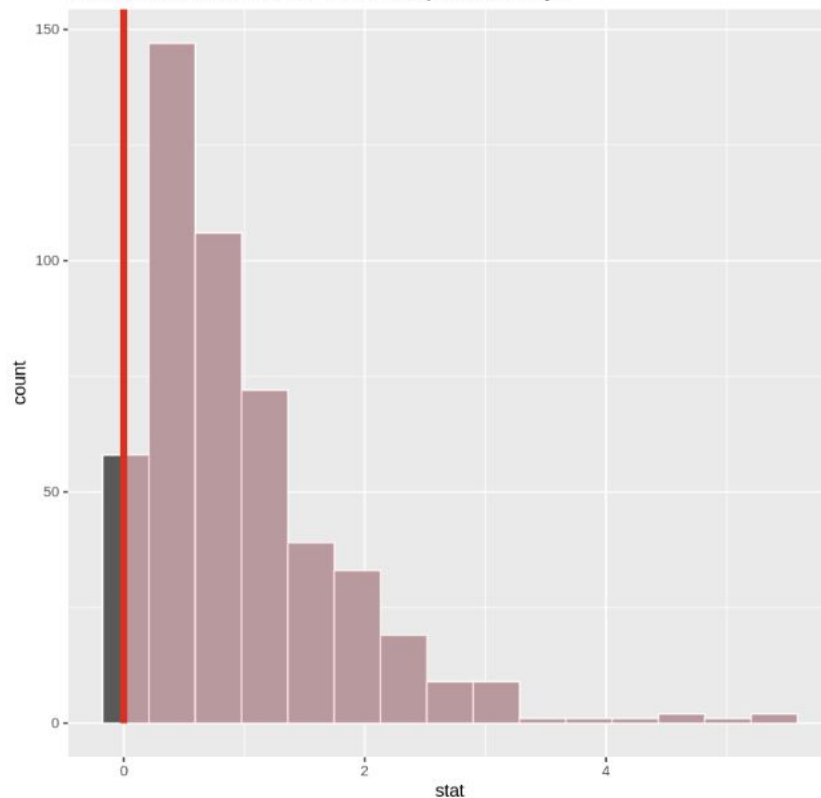
1

Permutation Test Summary

Permutation Test: Time of Day vs Departure Delays



Permutation Test: Time of Year vs Departure Delays



Discussion

- Key Takeaways
- Potential Expansion

Key Takeaways



Problem Statement:

What variable has the greatest effect on departure delays?

Suggested Solution:

Time of Day (Morning, Afternoon, Evening, Night)

Expansions

- Is this the same output for other Airlines?
- What about airport location?